Midterm Review

In Class Exercises

Exercise 1

The recurrence $T(n) = 2T(n-1) + 1$ has the solution $T(n) = O(2^n)$. Show that a substitution proof fails with the assumption $T(n) \leq c2^n$, where $c > 0$ is constant. Then show how to subtract a lower-order term to make a substitution proof work.

Exercise 1 (solution)

For the recurrence $T(n) = 2T(n-1) + 1$, if we use the guess that $T(n) \le c2^n$, the proof will fail:

$$\begin{aligned} T(n) &= 2T(n-1) + 1 \\ &\le 2(c2^{n-1}) + 1 \\ &= c2^n + 1 , \end{aligned}$$

which is greater than $c2^n$. Instead, subtract off a constant d, which we get to choose: $T(n) \le c2^n - d$. Now, we have

$$\begin{aligned} T(n) &= 2T(n-1) + 1 \\ &\le 2(c2^{n-1} - d) + 1 \\ &= c2^n - 2d + 1 , \end{aligned}$$

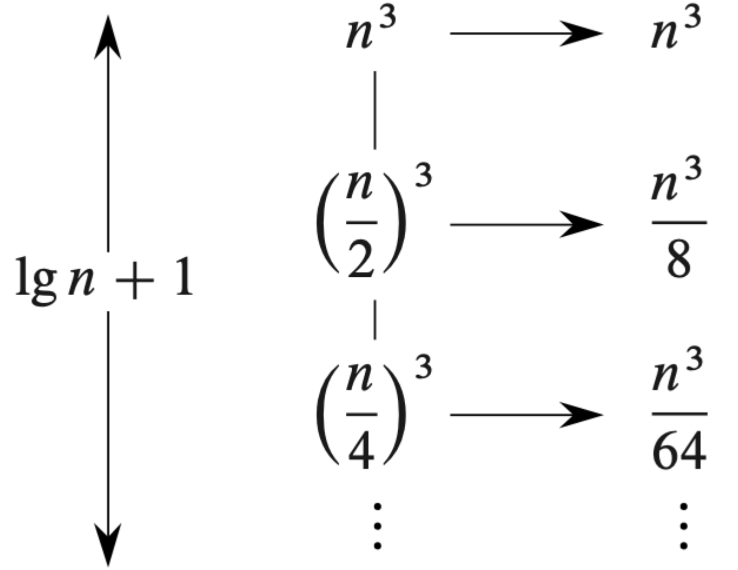
which is less than or equal to $c2^n - d$ if $d \ge 1$.

Exercise 2

- Sketch the recursion tree, and guess a good asymptotic bound on the solution of the recurrence below, then use the substitution method to verify your answer:

$$T(n) = T(n/2) + n^3.$$

Exercise 2 (solution)



$$\text{Total: } n^3 \sum_{k=0}^{\lg n} \frac{1}{8^k} < n^3 \sum_{k=0}^{\infty} \frac{1}{8^k} = O(n^3)$$

Exercise 2 (solution)

The recursion tree has a single node at each level, contributing $(n/2^k)^3 = n^3/8^k$ at each depth k. The total is

$$\begin{aligned}\sum_{k=0}^{\lg n} \frac{n^3}{8^k} &< n^3 \sum_{k=0}^{\infty} (1/8)^k \\ &= n^3 \cdot \frac{1}{1 - 1/8} \\ &= (8/7)n^3 \\ &= O(n^3) .\end{aligned}$$

Exercise 2 (solution)

Now, we prove that $T(n) = O(n^3)$ with the substitution method. We need to show that $T(n) \le cn^3$ for some constant c. We have

$$\begin{aligned} T(n) &= T(n/2) + n^3 \\ &\le c(n/2)^3 + n^3 \\ &= cn^3/8 + n^3 \\ &= n^3(c/8 + 1) , \end{aligned}$$

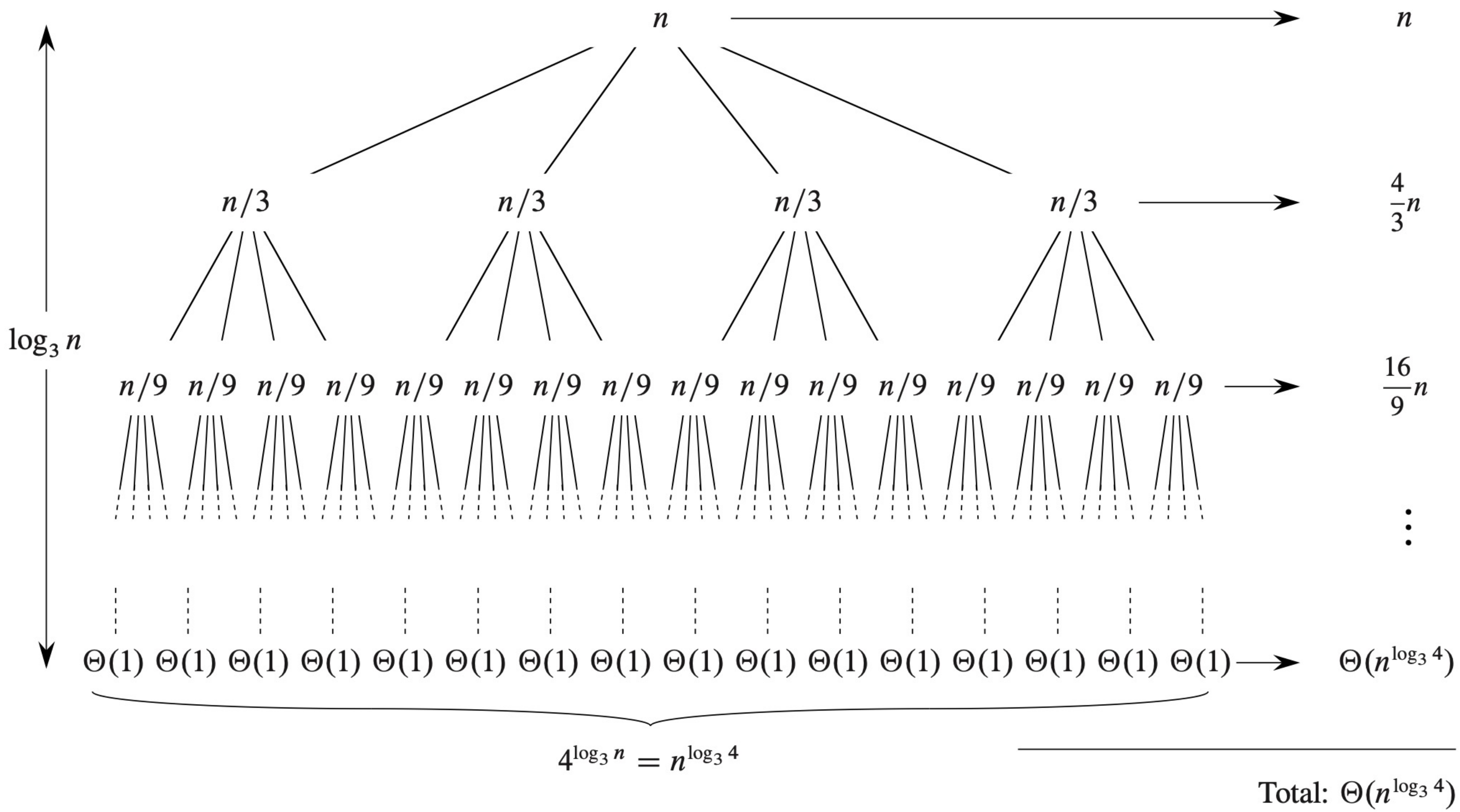
which is less than or equal to cn^3 if $c \ge 8/7$.

Exercise 3

- Sketch the recursion tree, and guess a good asymptotic bound on the solution of the recurrence below, then use the substitution method to verify your answer:

$$T(n) = 4T(n/3) + n.$$

Exercise 3 (solution)



Exercise 3 (solution)

$$\begin{aligned}
\sum_{k=0}^{\log_3 n-1} \left(\frac{4}{3}\right)^k n + \Theta(n^{\log_3 4}) &= n \cdot \frac{(4/3)^{\log_3 n} - 1}{(4/3) - 1} + \Theta(n^{\log_3 4}) \\
&< 3n(4/3)^{\log_3 n} + \Theta(n^{\log_3 4}) \\
&= 3n(4^{\log_3 n})(1/3)^{\log_3 n} + \Theta(n^{\log_3 4}) \\
&= 3n(n^{\log_3 4})(3^{-\log_3 n}) + \Theta(n^{\log_3 4}) \\
&= 3n^{\log_3 4+1} n^{-\log_3 3} + \Theta(n^{\log_3 4}) \\
&= 3n^{\log_3 4+1} n^{-1} + \Theta(n^{\log_3 4}) \\
&= 3n^{\log_3 4} + \Theta(n^{\log_3 4}) \\
&= \Theta(n^{\log_3 4}) .
\end{aligned}$$

Exercise 3 (solution)

Now, we prove that $T(n) = O(n^{\log_3 4})$ with the substitution method. We guess that $T(n) \le n^{\log_3 4} - cn$, where $c > 0$ is a constant that we will choose. We have

$$
\begin{aligned}
T(n) &= 4T(n/3) + n \\
&\le 4((n/3)^{\log_3 4} - cn/3) + n \\
&= 4n^{\log_3 4}(1/3)^{\log_3 4} - (4/3)cn + n \\
&= 4n^{\log_3 4}(3^{-\log_3 4}) - (4/3)cn + n \\
&= 4n^{\log_3 4}(4^{-\log_3 3}) - (4/3)cn + n \\
&= 4n^{\log_3 4}(4^{-1}) - (4/3)cn + n \\
&\le n^{\log_3 4} - cn
\end{aligned}
$$

if $-(4/3)cn + n \le -cn$ or, equivalently, $c \ge 3$. Hence, $T(n) = O(n^{\log_3 4})$.

Exercise 4

- Use the master method to give tight asymptotic bounds for the following recurrences:

$$T(n) = 2T(n/4) + 1.$$

$$2T(n/4) + n^2.$$

Exercise 4 (solution)

$$T(n) = 2T(n/4) + 1.$$

$T(n) = \Theta(\sqrt{n})$. Here, $f(n) = O(n^{1/2-\epsilon})$ for $\epsilon = 1/2$. Case 1 applies, and $T(n) = \Theta(n^{1/2}) = \Theta(\sqrt{n})$.

$$2T(n/4) + n^2.$$

$T(n) = \Theta(n^2)$. Now, $f(n) = n^2$, and so $f(n) = \Omega(n^{\log_b a+\epsilon})$ for $\epsilon = 3/2$. In order for case 3 to apply, we again have to check the regularity condition: $af(n/b) \le cf(n)$ for some constant $c < 1$. Here, $af(n/b) = n^2/8$, and so the regularity condition holds for $c = 1/8$. Therefore, case 3 applies.